

Online Bus Ticket Booking System Using Power BI

POWER BI PROJECT REPORT

Submitted by

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Abstract

The Online Bus Ticket Booking System using Power BI is designed to provide a streamlined solution for managing bus reservations, customer details, payments, and routes. By integrating data from Excel-based datasets (Bus Details, Customer Information, Bookings, and Payments), this system offers real-time insights through visualizations in Power BI. Users can analyze revenue, track seat availability, monitor payment status, and identify customer preferences. With intuitive dashboards and filters, businesses can optimize bus operations, forecast demand, and enhance decision-making. The system also supports automated data refreshes, ensuring that booking and payment data stays up to date, improving both operational efficiency and customer satisfaction.

Problem Statement

The current manual or semi-automated methods of managing bus ticket bookings are inefficient, prone to errors, and lack real-time visibility into key metrics like seat availability, revenue, and customer preferences. These inefficiencies result in missed opportunities for optimizing routes, managing demand, and improving customer service. Additionally, businesses face challenges in tracking payments, identifying unpaid bookings, and forecasting future demand. There is a need for a robust, data-driven solution that can automate the booking process, provide real-time insights, and allow stakeholders to make informed decisions. An integrated **Online Bus Ticket Booking System** using Power BI is proposed to solve these challenges by offering automated, centralized management of bookings, payments, and customer data with powerful analytics and visual reporting.

1. Bus Details Dashboard

Purpose

The Bus Details table is designed to manage payment transactions related to bus reservations. It tracks payment information to confirm bookings, ensure accurate financial records, and manage any issues related to payments for bus services.

Key Values

Payment_ID: A unique identifier for each payment transaction associated with a bus booking, allowing distinct tracking of every payment.

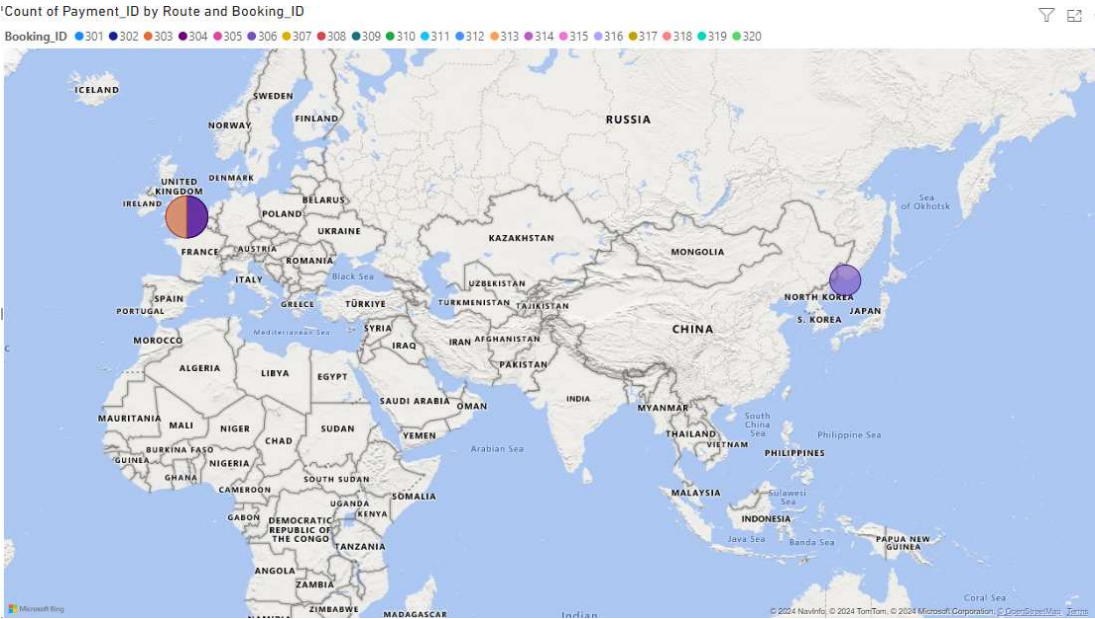
Booking_ID: The identifier for each bus booking to which a payment is associated. This linkage helps in managing which payment is for which specific bus reservation.

Payment_Date: The date when the payment for the bus booking was made. This helps in record-keeping, tracking timely payments, and maintaining chronological financial records.

Payment_Amount: The amount paid for the bus booking. This is crucial for ensuring that the correct amount is charged and paid per booking, supporting financial integrity.

Payment_Status: Indicates the status of the payment (e.g., completed, pending, refunded, or failed) for the bus booking, assisting in monitoring outstanding payments and handling payment-related issues.

Bus_ID	Sum of Seats_Booked	Sum of Total_Price
101	2	30
201	2	30
102	3	60
202	3	60
103	1	25
203	1	25
104	4	72
204	4	72
105	2	44
205	2	44
106	4	45
206	4	45
107	5	46
207	5	46
108	3	47
208	3	47
109	2	48
209	2	48
110	1	49
210	1	49
111	5	50
211	5	50
112	6	51
212	6	51
113	5	52
213	5	52
114	4	53
214	4	53
Total	69	1011



2. Customer Details Dashboard

Purpose

The Customer Details table is designed to store and manage key information about customers. It serves as a central repository for customer identification and contact information, which can be used for communication, tracking bookings or transactions, and customer relationship management.

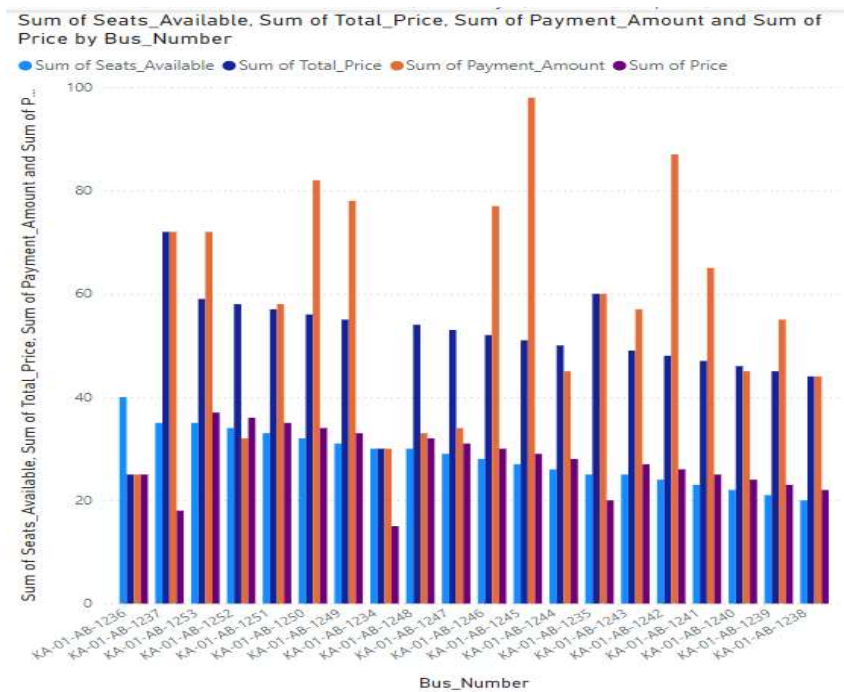
Key Values

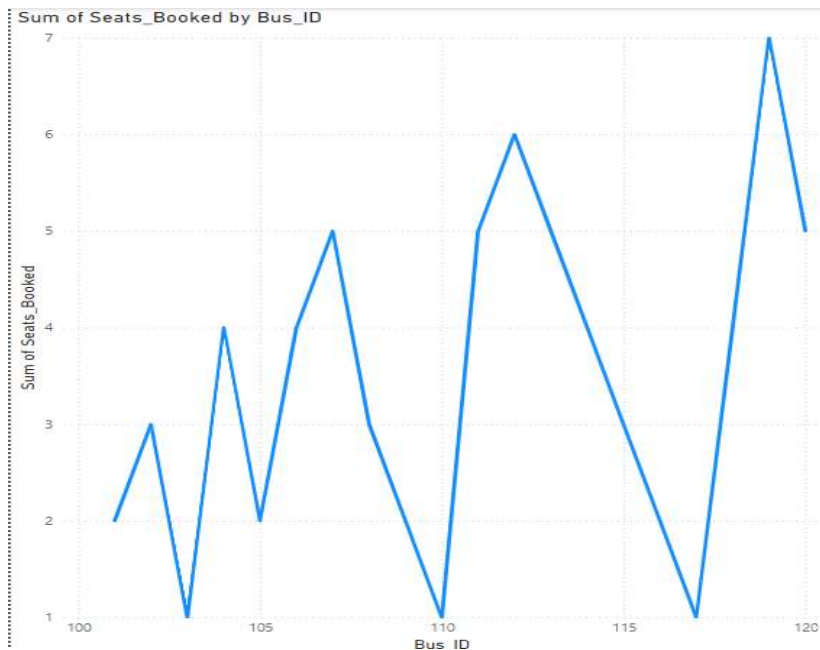
Customer_ID: A unique identifier for each customer. This ID is used to distinguish between customers and can be referenced in other related tables (such as bookings, payments, etc.) to link customer records to their transactions or interactions.

Name: The customer's full name, which helps in personalizing communication and identifying the customer in any correspondence or service interactions.

Email: The email address of the customer. This is used for sending booking confirmations, payment receipts, promotions, and any other customer service-related communications.

Phone: The customer's contact number, which can be used for real-time communication, such as customer support, booking confirmations, or urgent updates.





3. Booking Details Dashboard

Purpose

The Booking Details table is designed to manage and track information related to bus bookings. It links customers, buses, and transaction details to each booking, providing a clear record of each transaction, the number of seats booked, and the total price paid. This helps in organizing reservations, customer information, and financial transactions for the bus service.

Key Values

Booking_ID: A unique identifier for each booking. This ID is used to track individual bookings and associate them with related details such as customer information, bus details, and payment transactions.

Bus_ID: Identifies the bus for which the booking is made. This links the booking to a specific bus, helping in managing bus schedules, seat availability, and route planning.

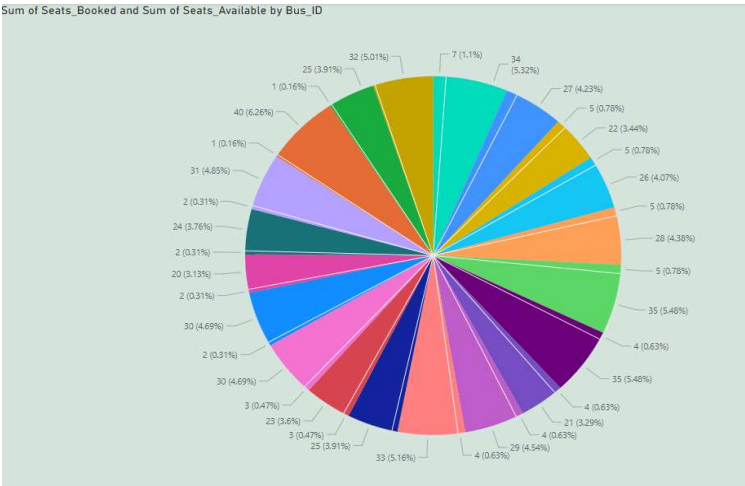
Customer_ID: The identifier for the customer who made the booking. This links the booking to a specific customer, allowing for customer relationship management and tracking customer-specific transactions.

Booking_Date: The date when the booking was made. This helps in organizing bookings by date, tracking booking trends, and managing schedules effectively.

Seats_Booked: The number of seats that have been reserved in the booking. This is essential for capacity management and ensuring that the bus's seating availability is

Total_Price: The total price paid for the booking, including any fees or discounts. This is crucial for financial tracking and ensuring that the correct amount has been charged for the booked seats.

Year	Quarter	Month	Day	Sum of Payment_Amount	Count of Customer_ID	Sum of Total_Price
2024	Qtr 4	October	1	₹ 30	1	₹ 30
2024	Qtr 4	October	2	₹ 60	1	₹ 60
2024	Qtr 4	October	3	₹ 25	1	₹ 25
2024	Qtr 4	October	4	₹ 72	1	₹ 72
2024	Qtr 4	October	5	₹ 44	1	₹ 44
2024	Qtr 4	October	6	₹ 55	1	₹ 45
2024	Qtr 4	October	7	₹ 45	1	₹ 46
2024	Qtr 4	October	8	₹ 65	1	₹ 47
2024	Qtr 4	October	9	₹ 87	1	₹ 48
2024	Qtr 4	October	10	₹ 57	1	₹ 49
2024	Qtr 4	October	11	₹ 45	1	₹ 50
2024	Qtr 4	October	12	₹ 98	1	₹ 51
2024	Qtr 4	October	13	₹ 77	1	₹ 52
2024	Qtr 4	October	14	₹ 34	1	₹ 53
2024	Qtr 4	October	15	₹ 33	1	₹ 54
2024	Qtr 4	October	16	₹ 78	1	₹ 55
2024	Qtr 4	October	17	₹ 82	1	₹ 56
2024	Qtr 4	October	18	₹ 58	1	₹ 57
2024	Qtr 4	October	19	₹ 32	1	₹ 58
2024	Qtr 4	October	20	₹ 72	1	₹ 59
Total				₹ 1,149	20	₹ 1,011



4. Payment Details Dashboard

Purpose

The Payment Details table is designed to manage and track payment transactions related to bookings. It helps monitor payments made for bus reservations, including payment status and the amount paid. This table is essential for financial tracking, verifying booking payments, and managing any payment-related issues, such as refunds or pending payments.

Key Values

Payment_ID: A unique identifier for each payment transaction. This ensures that every payment is individually tracked and can be referenced when needed (e.g., for reporting or issue resolution).

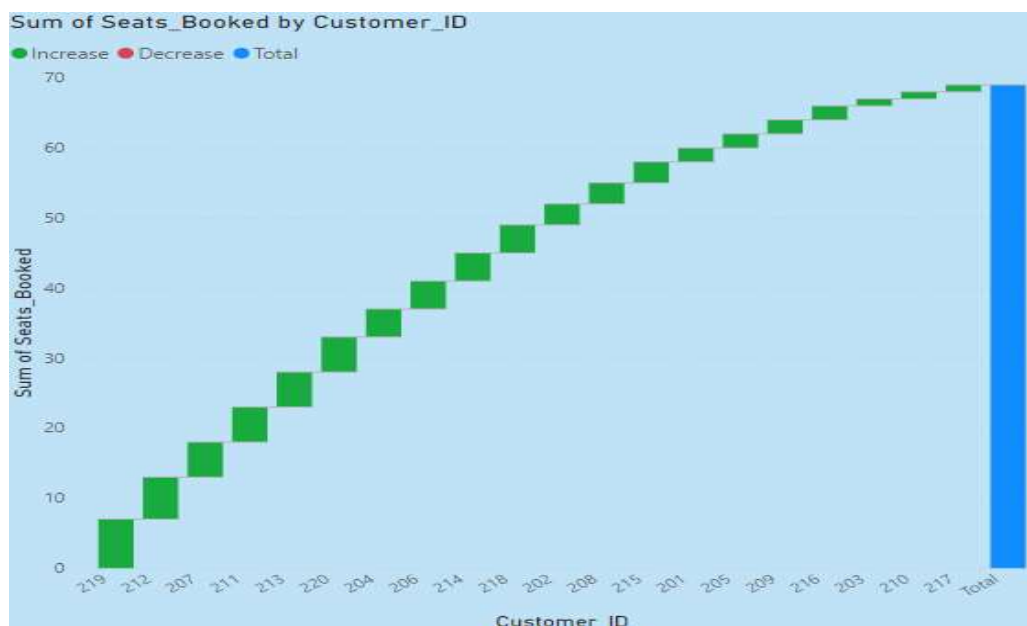
Booking_ID: The identifier linking the payment to a specific booking. This connection ensures that the payment corresponds to the right booking, helping to maintain accurate records of which bookings have been paid for.

Payment_Date: The date on which the payment was made. This is crucial for keeping track of when payments occur, supporting financial reporting, and ensuring timely transactions.

Payment_Amount: The total amount paid in the transaction. This value ensures that the correct amount has been received for the booking, allowing for accurate financial records and the management of outstanding balances or discrepancies.

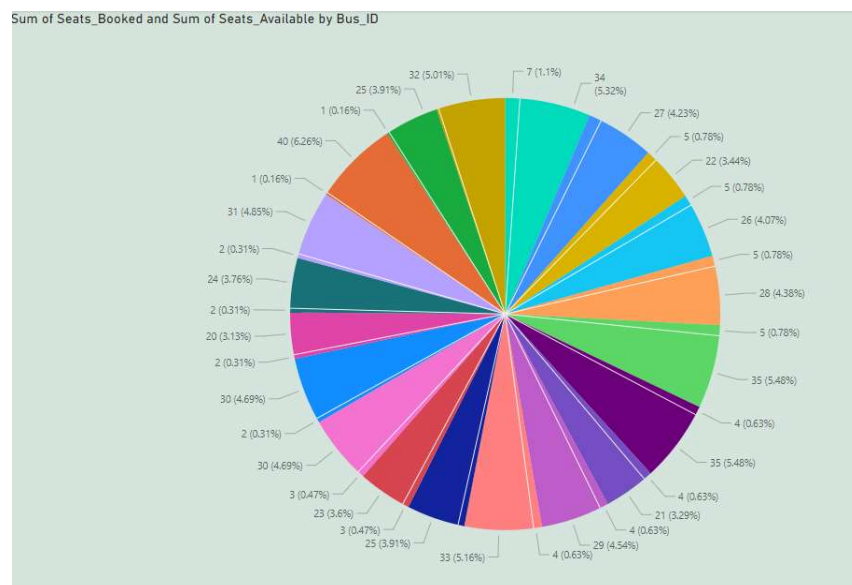
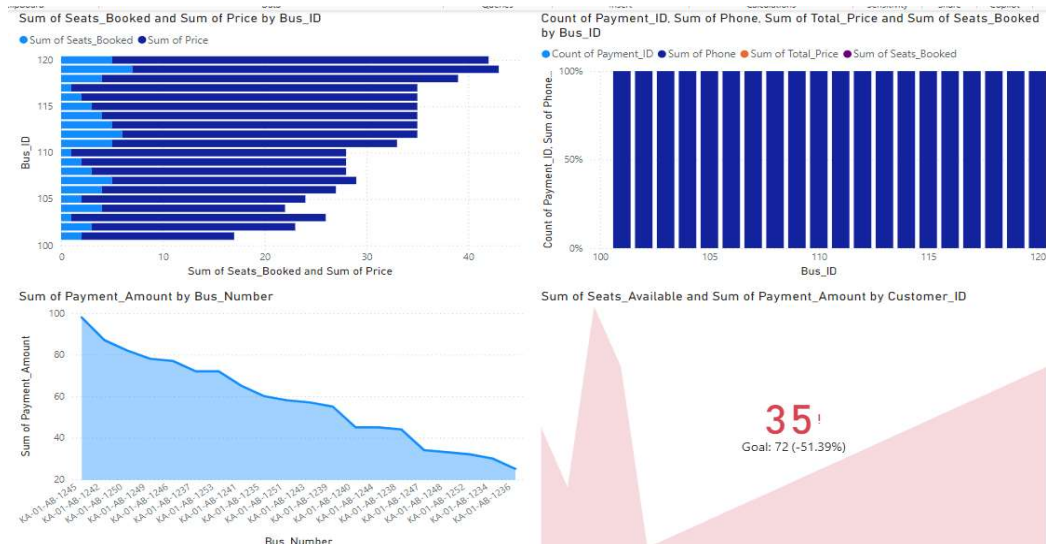
Payment_Status: Indicates the status of the payment (e.g., completed, pending, refunded, or failed). This status helps in monitoring outstanding payments and resolving issues related to unpaid or failed transactions.

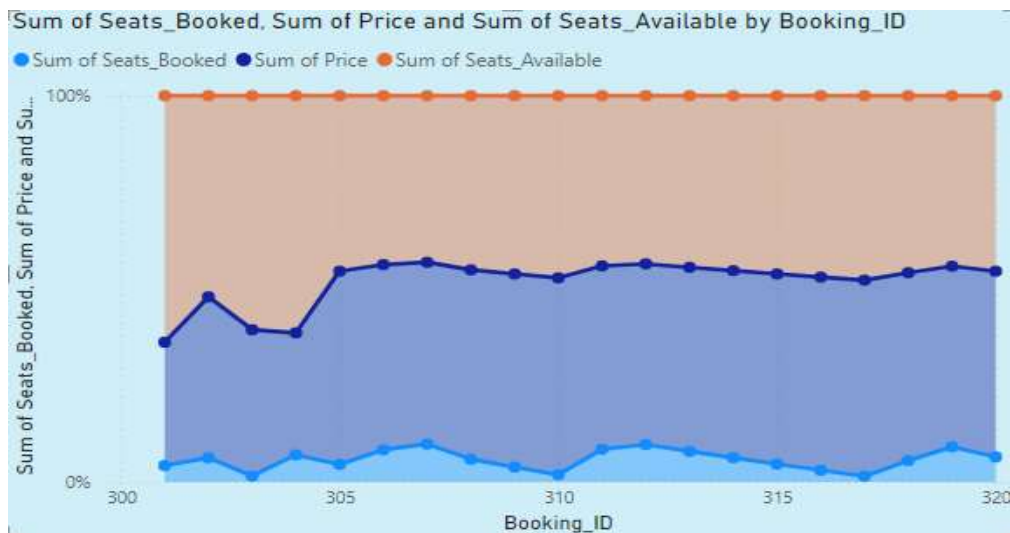
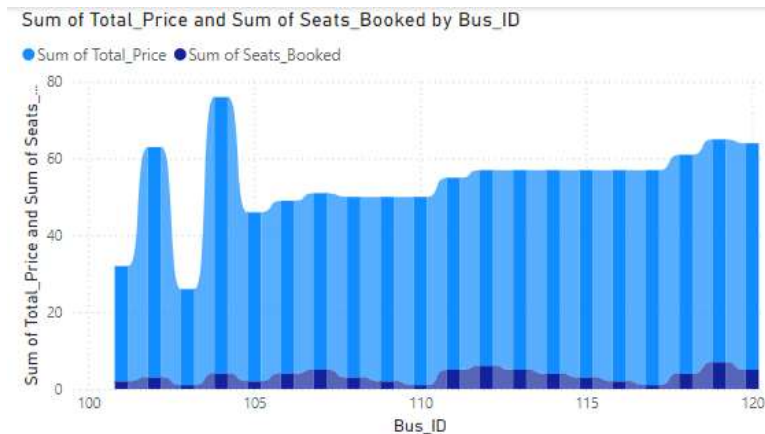
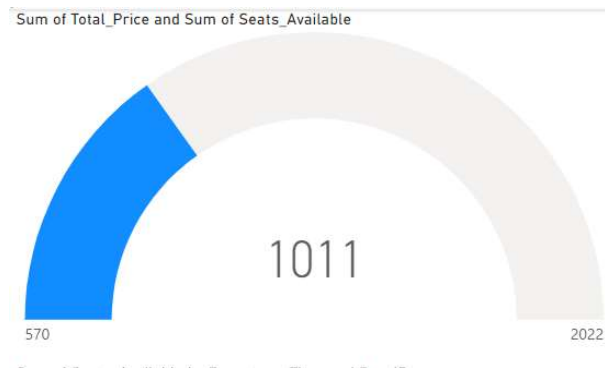
	Sum of Total_Price	Sum of Seats_Booked
	1011	69





Overall Dashboard Creation





Conclusion

The project aimed to develop a Bus Booking and Payment Management System that streamlines the process of booking bus services, managing customer data, tracking payments, and ensuring financial accuracy. By utilizing well-structured tables for customer details, booking details, and payment details, the system enables seamless operation, data management, and service delivery.